

Lnpt

OEE

A

P

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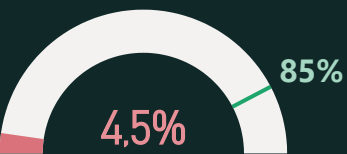
Date
Multi

Total Biscuits Produced
20.373.809

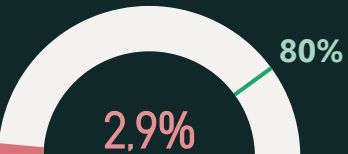
Good Biscuits Produced
13.552.673



OEE



TEEP



Biscuit Filling Machine 14.9%

Packaging Heat Machine 4.0%

Biscuit Boxing Machine 1.8%

Biscuit Sprinkling Machine 0.8%

Biscuit Pressing Machine 0.3%

Biscuit Mixing Machine 0.1%

Biscuit Topping Machine 0.0%

Biscuit Forming Machine 0.0%

Biscuit Jam Machine 0.0%

Biscuit Heating Machine 0.0%

Biscuit Filling Machine 11.6%

Packaging Heat Machine 3.6%

Biscuit Sprinkling Machine 0.8%

Biscuit Boxing Machine 0.4%

Biscuit Pressing Machine 0.3%

Biscuit Mixing Machine 0.0%

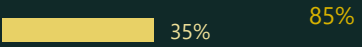
Biscuit Topping Machine 0.0%

Biscuit Jam Machine 0.0%

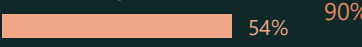
Biscuit Forming Machine 0.0%

Biscuit Heating Machine 0.0%

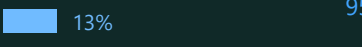
Capacity Utilization



Availability



Performance



Quality



Full Plant Time: 188,640

Effective Run Time

66.002

Down Time Loss

57.158

Planned Shut Down

67.061

Effective Run Time

66.002

Down Time Loss

57.158

Productive Time

8.346

Performance Loss

57.656

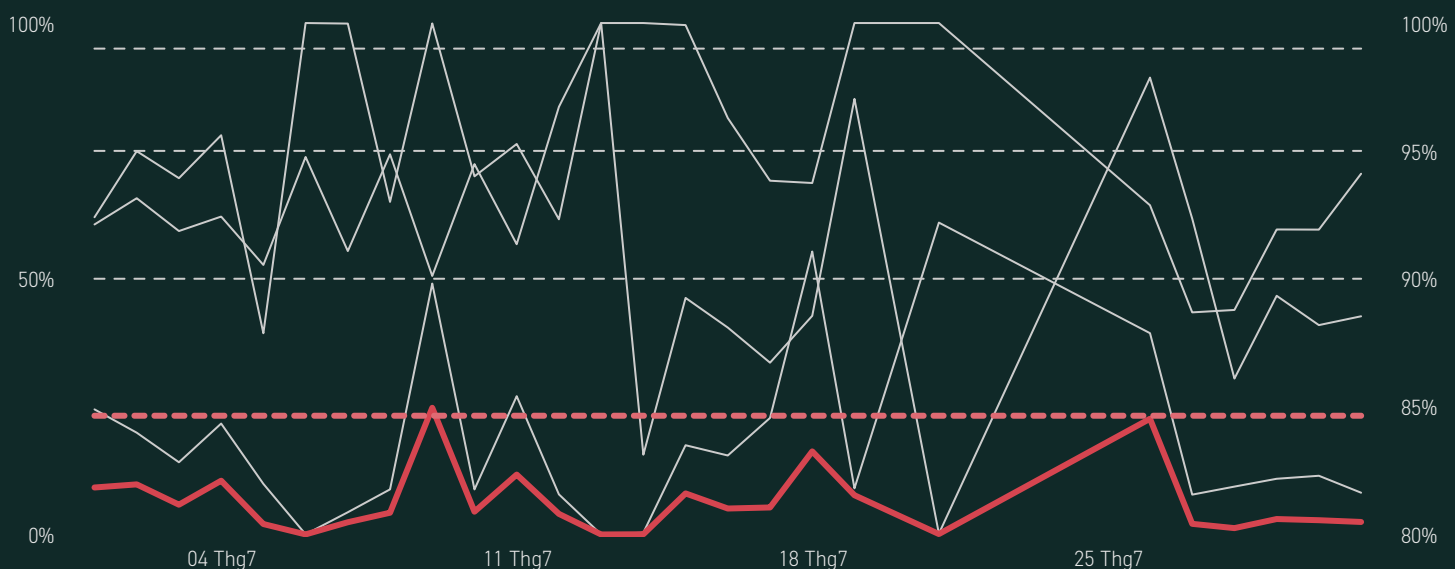
Fully Productive Time

5.552

Quality Loss

2.794

KPI by Day



Other



Lnpt

OEE

A

P

Q

Date
Multi

Total Plant
Time
188.640

Planned Run
Time
121.579

Effective Run
Time
66.002

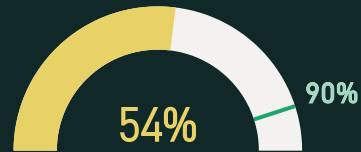
Down Time
Loss
57.158



Other



Availability



Biscuit Heating Machine 94%

Biscuit Topping Machine 91%

Biscuit Forming Machine 80%

Packaging Heat Machine 50%

Biscuit Filling Machine 46%

Biscuit Mixing Machine 45%

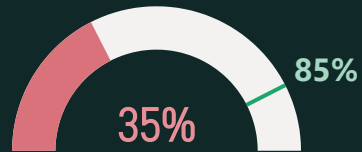
Biscuit Sprinkling Machine 44%

Biscuit Jam Machine 42%

Biscuit Pressing Machine 26%

Biscuit Boxing Machine 25%

Capacity Utilization



Biscuit Heating Machine 93%

Biscuit Topping Machine 91%

Biscuit Forming Machine 54%

Packaging Heat Machine 45%

Biscuit Sprinkling Machine 44%

Biscuit Jam Machine 42%

Biscuit Filling Machine 35%

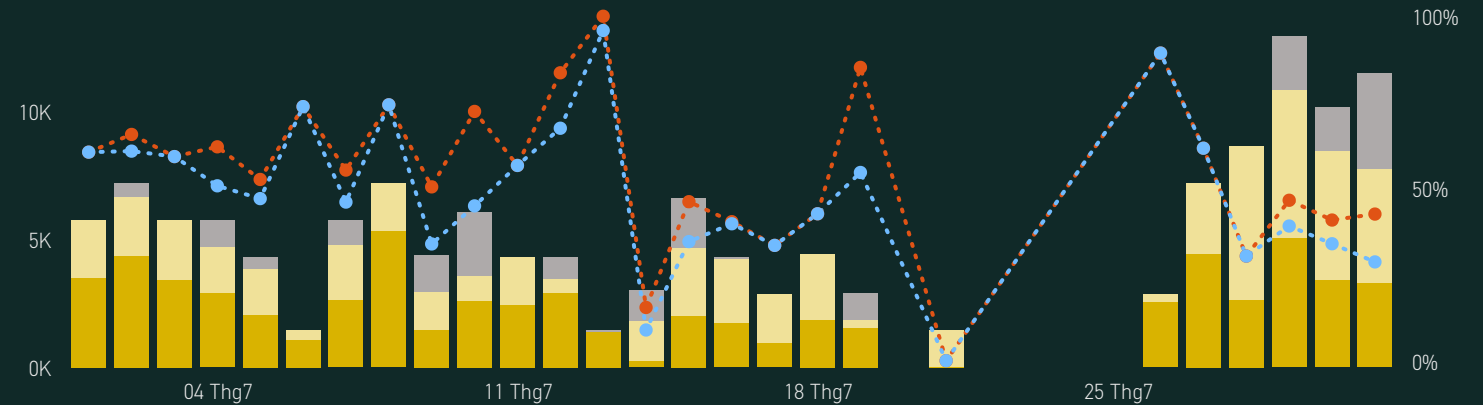
Biscuit Pressing Machine 26%

Biscuit Mixing Machine 24%

Biscuit Boxing Machine 6%

Availability by Day

Effective Runtime Down Time Loss Planned Shut Down % Availability % Capacity Utilization



Down Time Loss Category

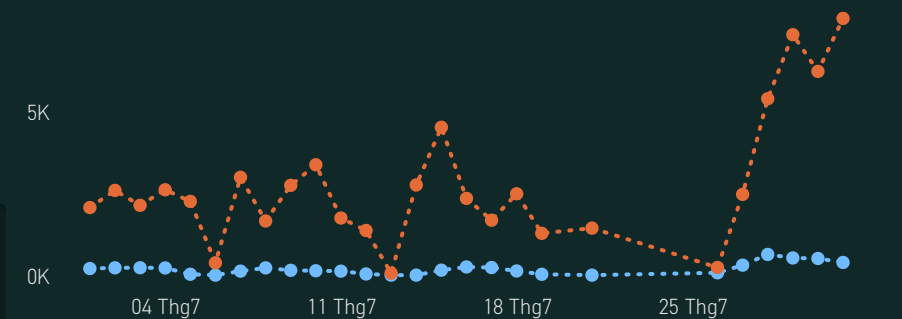
AL (Other Availability Loss) 37,516
CC (Changeover Cleaning) 19,642

Down Time Loss by Machine

Biscuit Filling Machine 17,818
Biscuit Mixing Machine 9,195
Biscuit Boxing Machine 7,854
Biscuit Pressing Machine 5,298
Biscuit Jam Machine 4,207

Minor and Major Stoppages

Time Minor Stoppage Time Major Stoppage



Minor Stoppage (stoppage with a duration under 3 minutes).
Major Stoppage (stoppage with a duration over 3 minutes).
Review Minor and Major Stoppage by Machine. Small gap between Duration of Minor and Major Stoppage indicates that Machine has many minor stoppages during the day, signifying poor machine performance.

Lnpt

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Date
Multi

Design
Performance
161M

Total
Produced
20M

Target Time
to Produce
8.346

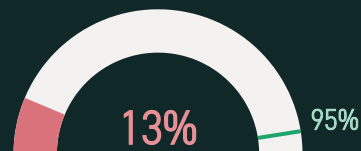
Performance
Loss (minutes)
57.656



Other



Performance



Biscuit Filling Machine 45%

Biscuit Sprinkling Machine 17%

Packaging Heat Machine 8%

Biscuit Boxing Machine 7%

Biscuit Pressing Machine 1%

Biscuit Mixing Machine 0%

Biscuit Jam Machine 0%

Biscuit Forming Machine 0%

Biscuit Topping Machine 0%

Biscuit Heating Machine 0%

Performance by Machine (minutes)

Performance Loss Total Produced

Biscuit Filling Machine	55%	45%
Biscuit Forming Machine	100%	
Biscuit Pressing Machine	99%	
Biscuit Jam Machine	100%	
Biscuit Mixing Machine	100%	
Biscuit Sprinkling Machine	83%	
Biscuit Boxing Machine	93%	
Biscuit Heating Machine	100%	
Packaging Heat Machine	92%	
Biscuit Topping Machine	100%	

Performance by Machine (biscuits)

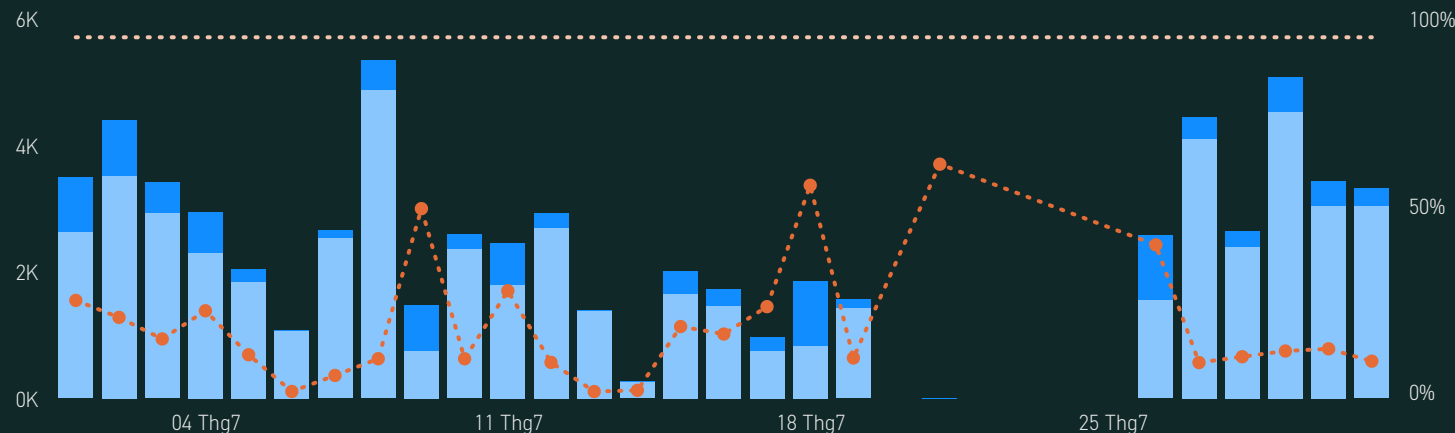
Performance Loss Total Produced

Biscuit Forming Machine	100%	
Biscuit Heating Machine	100%	
Biscuit Filling Machine	55%	45%
Biscuit Topping Machine	100%	
Biscuit Mixing Machine	100%	
Packaging Heat Machine	92%	
Biscuit Jam Machine	100%	
Biscuit Sprinkling Machine	83%	
Biscuit Boxing Machine	93%	
Biscuit Pressing Machine	99%	

Performance by Day (minutes)

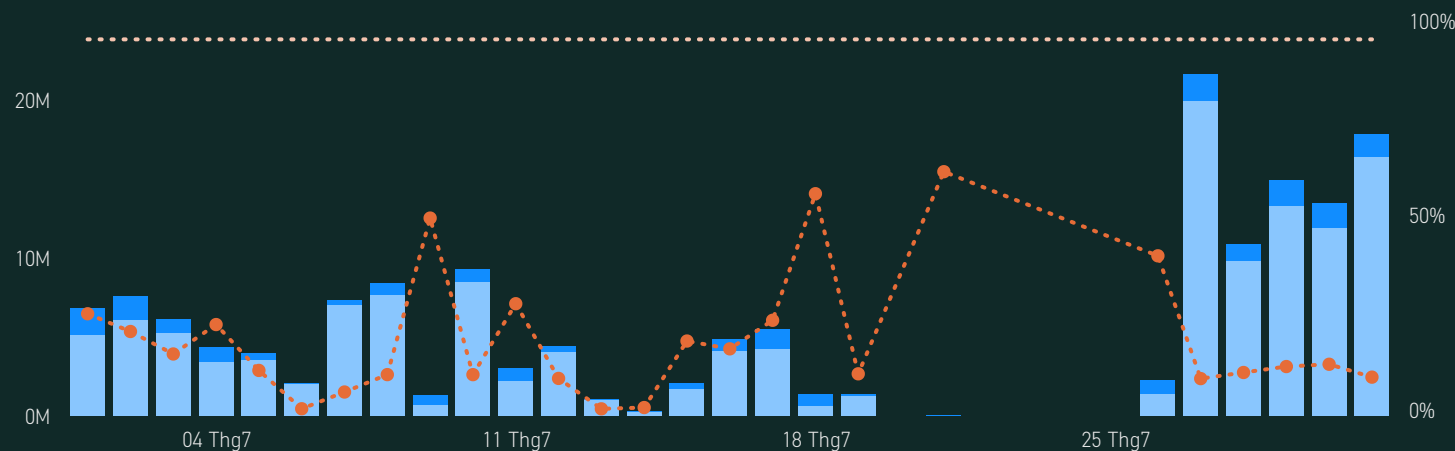
Target Time to Produce: time to produce an amount of biscuits based on Design Speed

Performance Loss Target Time to Produce % Performance % Performance Target



Performance by Day (biscuits)

Performance Loss (biscuits) Total Produced % Performance % Performance Target



Lnpt

OEE

A

P

Q

Date
Multi

Total Biscuits
Produced
20M

Good Biscuits
Produced
14M

Target Time
to Produce
5.552

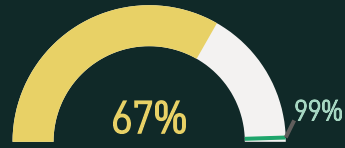
Quality Loss
(minutes)
2.794



Other



Quality



Biscuit Boxing Machine 100%

Biscuit Forming Machine 100%

Biscuit Heating Machine 100%

Biscuit Jam Machine 100%

Biscuit Mixing Machine 100%

Biscuit Pressing Machine 100%

Biscuit Topping Machine 100%

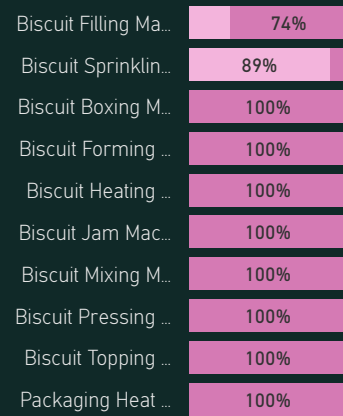
Packaging Heat Machine 100%

Biscuit Filling Machine 74%

Biscuit Sprinkling Machine 11%

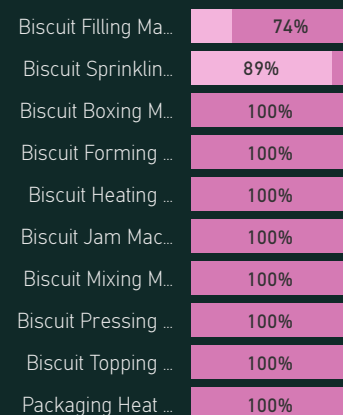
Quality by Machine (minutes)

Quality Loss (bis... Good Produc...



Quality by Machine (biscuits)

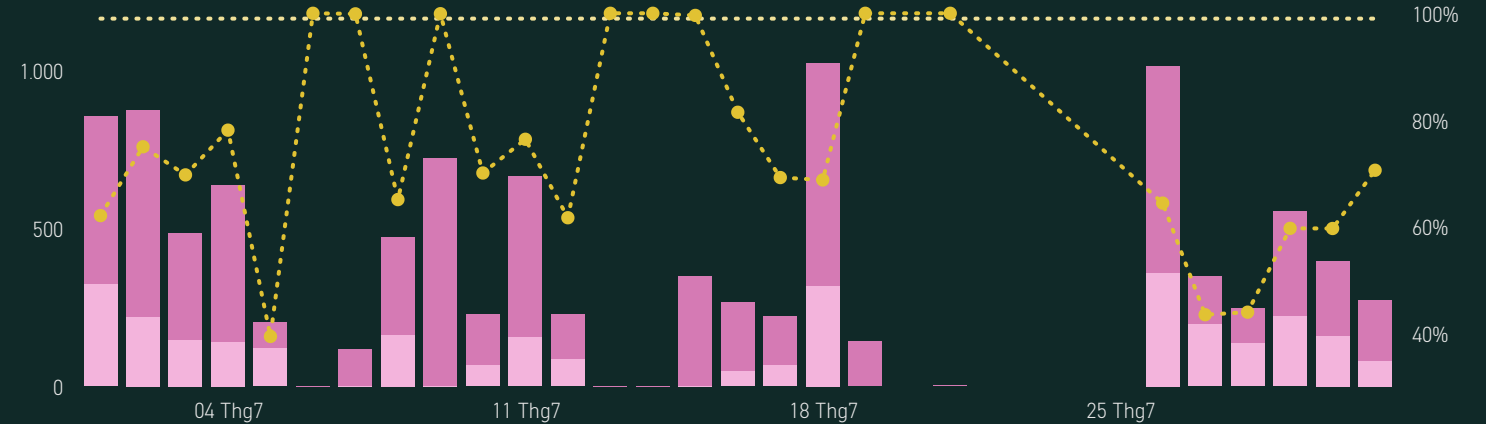
Quality Loss (biscuit) Target Time to Pro...



Quality by Day (minutes)

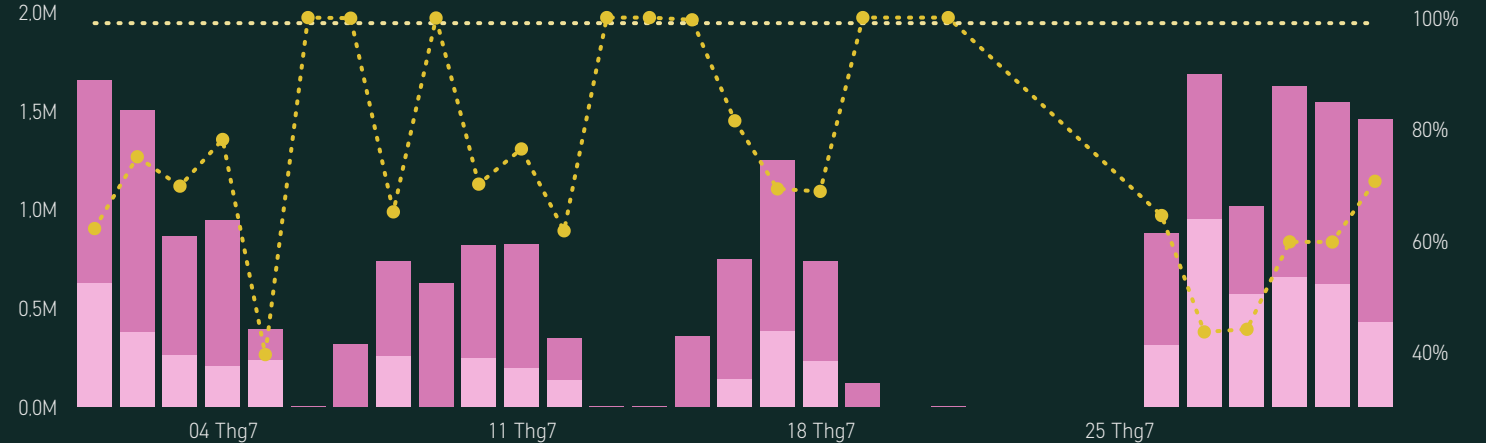
Target Time to Produce: time to produce an amount of biscuits based on Design Speed

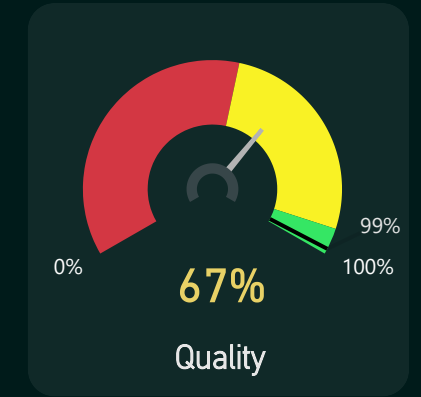
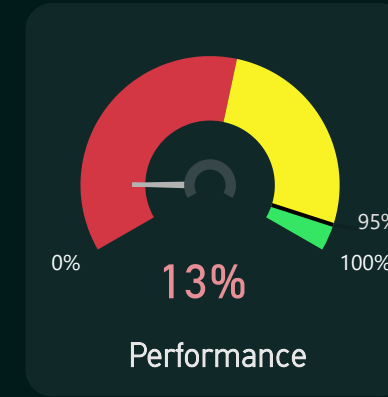
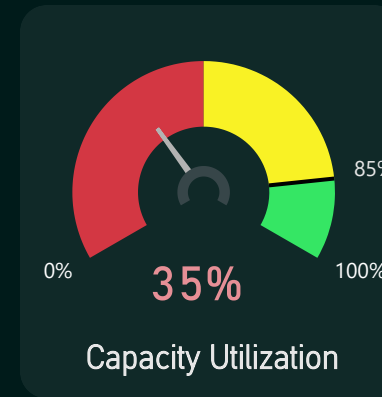
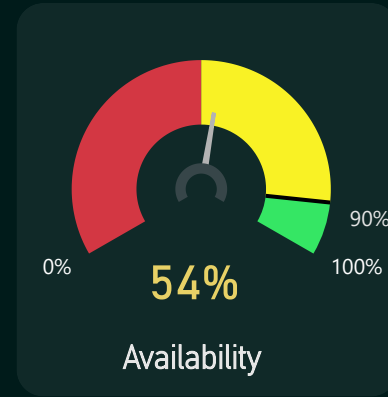
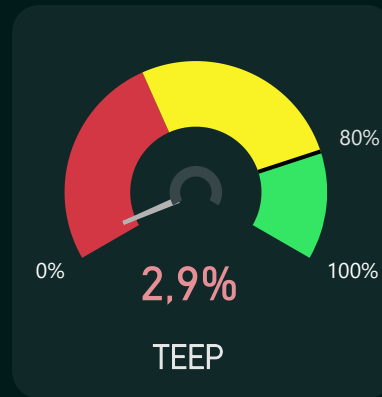
Quality Loss Target Time to Produce % Quality % Quality Target



Quality by Day (biscuits)

Quality Loss (biscuit) Good Produced % Quality % Quality Target





Fill - Mix - Form - Heat

- Top | Press - Jam - Sprinkle

Packaging Heat - Box

Biscuit Filling Machine	
Throughput (pcs per min)	1.052
Design Speed (pcs per min)	2.364
Average Changeover (mins)	5,39
Mean-time-to-repair (MTTR)	3,80
Mean-time-between-failures (MTBF)	3,82
Total Produced	16M
Total Down Time	17.818
OEE	14,9%
% Availability	46%
% Performance	45%
% Quality	74%

Biscuit Mixing Machine	
Throughput (pcs per min)	4
Design Speed (pcs per min)	2.042
Average Changeover (mins)	4,56
Mean-time-to-repair (MTTR)	88,09
Mean-time-between-failures (MTBF)	37,36
Total Produced	31K
Total Down Time	9.195
OEE	0,1%
% Availability	45%
% Performance	0,2%
% Quality	100%

Biscuit Forming Machine	
Throughput (pcs per min)	1
Design Speed (pcs per min)	1.602
Average Changeover (mins)	74,34
Mean-time-to-repair (MTTR)	160,98
Mean-time-between-failures (MTBF)	355,33
Total Produced	9K
Total Down Time	3.406
OEE	0,04%
% Availability	80%
% Performance	0,05%
% Quality	100%

Biscuit Heating Machine	
Throughput (pcs per min)	0,2
Design Speed (pcs per min)	958
Average Changeover (mins)	79,54
Mean-time-to-repair (MTTR)	349,39
Mean-time-between-failures (MTBF)	2.355,70
Total Produced	2K
Total Down Time	588
OEE	0,02%
% Availability	94%
% Performance	0,02%
% Quality	100%